DELIVERABLE D2.3

Open Science Activities Report

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SEMDR

Smart Energy Management and Demand Response

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Abstract

This report documents the open science activities undertaken within the SEMDR project (Smart Energy Management & Demand Response Capabilities for Sustainable Industry) over its 12-month duration (April 2025 - March 2026). In alignment with the Cyprus Open Science Practices framework and the European Commission's Open Science Policy, the consortium has implemented a comprehensive approach to ensuring that the project's research outputs including publications, software, datasets, and methodological documentation, are made as openly accessible as possible while protecting commercially sensitive information.

The project has delivered a substantial set of concrete open science outputs. The consortium has published three open-source repositories through a dedicated SEMDR project GitHub organisation (https://github.com/semdr-project), covering Cyprus day-ahead electricity price forecasting tools, a combined weather and electricity price dataset, and an EnergyPlus building thermal simulation baseline dataset for Mediterranean hotel HVAC research. Each repository has been integrated with Zenodo to provide persistent DOI-based citation and long-term preservation. A peer-reviewed conference paper was presented at the 32nd Scientific Conference of EU CIGRE in Athens in December 2025 and published in the conference proceedings. All public deliverables have been made available through the project website (https://semdrcy.replit.app), which has been operational since M3 (June 2025).

The consortium has identified nine distinct open science outputs (OS-1 through OS-9), of which six are already completed or operational at the time of writing. The remaining outputs - a methods journal paper, an anonymised pool pump production dataset, and an anonymised HVAC production dataset - are committed or planned for the post-project period, with a firm September 2026 deadline on the pool pump dataset. All activities are designed to ensure full compliance with the FAIR data principles as outlined in the project's Data Management Plan (D2.4).

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1 Introduction

This section introduces the purpose and scope of the Open Science Activities Report, describes the policy framework guiding the consortium's approach, and establishes the relationship between this deliverable and the companion documents that together form the SEMDR project's open science and data management framework.

1.1 Purpose and Scope

This deliverable (D2.3) covers all activities related to open access publishing, open data sharing, open-source software, and adherence to the FAIR data principles throughout the project lifecycle and into the post-project period.

The SEMDR project recognises the importance of open science in maximising the impact of publicly funded research. By making the project's findings, data, and methodologies accessible to the wider research and industry communities, the consortium aims to enable further innovation in smart energy management and demand response for buildings, while also supporting the broader European energy transition. At the same time, the project operates in a commercial context where certain algorithmic and operational details constitute the intellectual property of the consortium partners. This report documents how the consortium balances these considerations, following the principle of "as open as possible, as closed as necessary."

1.2 Relationship to Other Deliverables

This Open Science Activities Report is closely linked to several companion deliverables that together define the project's approach to dissemination, exploitation, and data management:

- **D2.1 (Dissemination, Communication, and Outreach Activities Report):** Provides the full context for the project's conference presentations and planned journal publications, including author details, venues, and audience engagement outcomes.
- **D2.2 (Commercialisation Plan):** Documents the IPR framework that defines the boundary between open and proprietary outputs, including the background and foreground IP register and the provisional IP ownership split between consortium partners.
- **D2.4 (Data Management Plan):** Contains the detailed operational specifications for data handling, storage, security, FAIR compliance, metadata standards, and long-term preservation strategies.
- **D1.3 (Interim Progress Report):** Provides the authoritative timeline baseline for all project activities referenced in this report.
- **D5.1 (System Design and Development Report):** Describes the system architecture and data pipeline that determines what data is collected and how it flows through the platform.

Where this report provides a high-level overview of open science activities and achievements, D2.4 contains the detailed technical specifications for data formats, metadata standards, access procedures, and preservation strategies.

2 Open Access Publications

The SEMDR publication strategy aims to disseminate the project's scientific and technical contributions through peer-reviewed venues while ensuring open access to all resulting publications. This section describes the completed and planned publications arising from the project.

2.1 Publication Strategy

The consortium's publication strategy prioritises conference presentations and papers at recognised international venues to establish scientific visibility and peer validation, journal articles in open access or hybrid journals to provide detailed technical contributions, and open publication of all public project deliverables through the project website. All publications resulting from the SEMDR project acknowledge the RIF and EU NextGenerationEU funding and are made available in open access format either through gold open access (publication in an open access journal) or green open access (deposit of the accepted manuscript in an open repository such as Zenodo).

2.2 Publications Inventory

Table 1 provides a summary of all publications produced or planned within the SEMDR project. The full dissemination context for each publication, including audience engagement and impact, is documented in D2.1.

ID	Title / Description	Type	Status	Venue
P-01	Real-World Demonstration of Optimized Pool Pump Control for Demand Response in the Hospitality Sector	Conference paper	Published (Dec 2025)	32nd EU CIGRE, Athens
P-02	Validated Energy Savings Through Smart Demand Response in the Hospitality Sector: A Pilot Study in Cyprus	Journal article	In preparation	Applied Energy / Energy and Buildings (Elsevier)

Table 1 - Publications Inventory

2.3 CIGRE Conference Paper (P-01)

The SEMDR project's first formal publication was a conference paper presented at the 32nd Scientific Conference of EU CIGRE in Athens, Greece, in December 2025. The paper, entitled "Real-World Demonstration of Optimized Pool Pump Control for Demand Response in the Hospitality Sector," was presented by Dr. Andrey Sesyuk (Project Coordinator) on behalf of the consortium.

The paper presents real-world performance results from the pool pump demand response system, including energy cost reductions achieved through intelligent scheduling aligned with electricity tariff structures and demand response signals. The pool pump scheduling algorithm (SimplePumpScheduler) optimises pump operation timing based on day-ahead electricity prices

from the Transmission System Operator of Cyprus (TSOC), while maintaining water quality within safe thresholds as measured by ORP, pH, and temperature sensors. The publication demonstrates the practical applicability of the SEMDR approach in the hospitality sector and provides a foundation for further research dissemination.

In accordance with the project's open science commitments, the CIGRE conference proceedings will be deposited on Zenodo with a persistent DOI within six months of the proceedings becoming available, ensuring long-term accessibility and citability. The full dissemination details for this activity, including audience engagement and photographs, are documented in D2.1 (Section 3.2.2).

2.4 Planned Journal Paper (P-02)

A comprehensive journal article is in preparation for submission to a peer-reviewed energy journal (primary targets: Applied Energy or Energy and Buildings, both published by Elsevier) in Q2 2026. The article will present the complete SEMDR methodology across the demonstration systems such as HVAC (heating, ventilation, and air conditioning) and pool pump, along with the full pilot deployment results from the pilot site.

The article will include a detailed description of the project's optimisation algorithms at a general technical level (without disclosing proprietary implementation details), comprehensive experimental results across both energy systems, cost-benefit analysis of the demand response approach for the hospitality sector, and recommendations for deployment in similar buildings. The article will be submitted to an open access or hybrid journal to ensure compliance with the project's open access commitments. A pre-print will be deposited on Zenodo upon submission to ensure immediate open access. The full publication plan for P-02 is described in D2.1 (Section 3.3.1).

3 Open-Source Software and Datasets

A central achievement of the SEMDR project's open science activities is the publication of three open-source repositories covering software and datasets of interest to the Cyprus and Mediterranean energy research community. This section describes the dedicated SEMDR project GitHub organisation, the three published repositories, and the Zenodo integration that provides persistent citation and long-term preservation.

3.1 SEMDR Project GitHub Organisation

The consortium established a dedicated GitHub organisation for the SEMDR project at https://github.com/semdr-project. The decision to create a project-specific organisation, rather than host the repositories under a consortium partner's account, reflects the consortium's intent to present the open science outputs as deliverables of the SEMDR project rather than as commercial assets of any individual partner. This arrangement also ensures that the boundary between the project's open-source outputs and proprietary commercial platform is unambiguous in the public record.

The organisation hosts a landing repository (*semdr-project/semdr-open-science*) that provides a single entry point to all project open science outputs, with links to each of the published repositories, the CIGRE conference paper, the project website, and the planned future Zenodo deposits. Each repository within the organisation carries a consistent funding acknowledgement identifying the SEMDR project (COM-CONCEPT-ENERGY/0624/0160) and the co-funding by the European Union through NextGenerationEU and the Research and Innovation Foundation.

3.2 Published Repositories

Table 2 summarises the three open-source repositories published through the SEMDR project GitHub organisation. Each repository has been integrated with Zenodo via GitHub's native integration, which automatically mints a persistent DOI for each tagged release and ensures the repository contents are preserved in CERN's long-term archival infrastructure.

ID	Repository	Licence	Type	Description
OS-1	semdr-project/cyprus-energy-market-prices-forecaster	MIT	Software	Cyprus day-ahead electricity price forecasting tools. Includes data acquisition scripts and forecasting models for research and analytical use.
OS-2	semdr-project/cyprus-weather-electricity-prices	MIT (code) / CC BY 4.0 (data)	Software and dataset	Combined weather and day-ahead electricity price dataset for Cyprus, with acquisition and preparation scripts. Public market and weather data restructured for research use.
OS-3	semdr-project/limassol-hotel-hvac-energyplus-dataset	CC BY 4.0	Dataset	Five fixed-setpoint EnergyPlus simulation baselines (12, 14, 16, 18, 20°C chilled water supply) for a Mediterranean large hotel building, based on the publicly available US Department of Energy reference building model. Intended as a benchmark for researchers developing their own HVAC control strategies.

Table 2 - Published Open-Source Repositories

3.3 OS-1: Cyprus Electricity Price Forecaster

This repository provides open-source tools for Cyprus day-ahead electricity price forecasting. It includes data acquisition scripts that retrieve public market data from the Transmission System Operator of Cyprus (TSOC), along with forecasting models and supporting documentation. The repository has been available throughout the project period under an MIT licence.Both licences satisfy the project's open science objectives,with the MIT choice reflecting a preference for lower adoption friction among potential users.

3.4 OS-2: Cyprus Weather and Electricity Prices Dataset

This repository combines Cyprus day-ahead electricity prices with corresponding weather observations, providing a time-aligned dataset suitable for research on the relationship between weather conditions, electricity demand, and market pricing. The repository also includes the data acquisition pipeline used to construct the dataset, enabling other researchers to extend the dataset with more recent data. All data in this repository is derived from publicly available sources, and the dataset has been reviewed to ensure no commercially sensitive or proprietary information is included.

3.5 OS-3: Limassol Hotel HVAC EnergyPlus Dataset

This repository publishes five fixed-setpoint EnergyPlus simulation baselines for a Mediterranean large hotel building, built on the publicly available US Department of Energy reference building model. The five baselines correspond to chilled water supply temperatures of 12, 14, 16, 18, and 20°C, providing a set of comparable operating points against which researchers can evaluate their own HVAC control strategies. The dataset is positioned as a benchmark for the research community - a common reference point for controller development and comparison.

The dataset contains only simulation output from the unmodified reference building at fixed setpoints. It does not include any calibrated building model derived from the demonstration hotel, nor any output from the project's control optimisation algorithms. This scoping decision was deliberate: the consortium sought to release a genuinely useful research dataset while preserving the proprietary calibration and control logic that constitute the commercial intellectual property. The repository README explicitly invites researchers to benchmark their own controllers against these fixed-setpoint baselines.

3.6 Zenodo Integration and Persistent Identifiers

Each of the three repositories has been integrated with Zenodo via GitHub's native Zenodo integration. Tagging a release on GitHub automatically creates a corresponding Zenodo deposit with a persistent DOI, links the deposit to the GitHub record, and ensures the contents are preserved in CERN's archival infrastructure regardless of any future changes to the GitHub hosting arrangements. Each deposit is associated with the SEMDR community on Zenodo, which provides a coherent, discoverable collection of all project outputs.

The DOI associated with each repository release will be surfaced on the landing repository (*semdr-project/semdr-open-science*) and on the project website, providing a single canonical reference point that researchers can cite. Subsequent releases will receive new DOIs, with the Zenodo version history preserving the relationship between releases.

4 Open Data: Production Datasets

In addition to the open-source repositories described in Section 3, the consortium has committed to publishing anonymised production datasets derived from the live pilot deployment at the demonstration hotel. These datasets are distinct from the simulation baselines published in OS-3: they represent real operational data from the deployed system and are of value to the research community precisely because such real-world datasets are rare in the building energy literature. This section describes the datasets committed for release, the anonymisation approach, and the data that is excluded from open publication.

4.1 Datasets for Open Publication

Two anonymised production datasets are committed for release following the project period. These are derived from the operational data collected at the pilot deployment.

- **Anonymised pool pump energy and water quality time-series (OS-5):** Energy consumption readings (kW, kWh), pump operational status, and water quality parameters (ORP, pH, temperature) from the pool pump deployment. The dataset will be accompanied by a comprehensive data dictionary and quality notes, and will have all location identifiers removed. This dataset is the project's primary committed open data output, with a firm September 2026 release deadline.
- **Anonymised HVAC energy consumption time-series (OS-7):** Real production energy consumption data from the live HVAC demand response deployment. This dataset is distinct from the EnergyPlus simulation baselines in OS-3: where OS-3 provides synthetic data from the unmodified reference building, OS-7 will capture real production behaviour from the live algorithm. Release is planned following the completion of the HVAC cooling season validation (approximately late 2026).

4.2 Anonymisation Approach

Anonymisation of production datasets focuses on removing any information that could identify the demonstration hotel or reveal commercially sensitive operational details. Specific measures include removal of the hotel name, GPS coordinates, and street address from all metadata; removal of any fields that could identify specific equipment model numbers tied to the hotel's installed base; and aggregation or masking of patterns that could reveal occupancy-linked operational schedules. The anonymisation approach is designed to preserve the scientific value of the data - the time-series structure, the relationships between variables, the physical realism of the measurements - while preventing identification of the site or disclosure of information the hotel considers commercially confidential.

Datasets will be released under the Creative Commons Attribution 4.0 International (CC BY 4.0) licence upon publication, allowing for maximum reusability while ensuring appropriate attribution to the SEMDR project.

4.3 Data Exclusions

Certain categories of data generated within the SEMDR project are excluded from open publication to protect legitimate commercial and privacy interests. The boundary between open and proprietary outputs is documented in the IPR framework in D2.2; this section summarises the principal exclusions:

- **Proprietary algorithm parameters:** The specific implementation details of the demand response algorithms are maintained as confidential intellectual property. This includes the SimplePumpScheduler pricing logic, the dynamic programming optimizer state-space structure, the LightGBM feature set and model weights, cost function weights, and guardrail threshold calibration methodology. The EnergyPlus dataset published in OS-3 contains only fixed-setpoint baselines from the unmodified public reference building, not the project's calibrated building model or any optimiser output.
- **Detailed site-specific operational data:** Granular data that could reveal the demonstration hotel's commercial energy costs, precise operational schedules, or specific equipment configurations is not released in identifiable form. Anonymised production datasets (OS-5, OS-7) are released in a form that preserves scientific value without exposing commercially sensitive detail.
- **AccuWeather raw data:** Outdoor weather data consumed from the AccuWeather API is excluded from open publication due to the API's terms of service, which restrict redistribution. The weather data used in OS-2 is sourced from different, openly redistributable data feeds.
- **Personal data:** The SEMDR system does not collect guest or staff personal data. No personal data exclusions are necessary.

Raw and processed data is stored on AWS cloud infrastructure (see D5.1 for the full system architecture), ensuring secure backup and accessibility for the consortium during the project period. Data is maintained in standardised formats (CSV for time-series data, JSON for structured data, IDF for EnergyPlus models) with comprehensive metadata following the Dublin Core standard.

5 FAIR Data Compliance

The FAIR principles (Findable, Accessible, Interoperable, Reusable) are central to the SEMDR project's data management approach. This section summarises the implementation of each principle across the project's open data outputs. The detailed technical specifications for FAIR compliance are documented in D2.4.

5.1 FAIR Compliance Summary

Table 3 summarises how the SEMDR project implements each FAIR principle across the open-source repositories and datasets.

Principle	Requirement	SEMDR Implementation
Findable	Persistent identifiers, rich metadata, registered in searchable resources	All published repositories carry persistent Zenodo DOIs via GitHub-Zenodo integration. Dublin Core metadata accompanies each deposit. The SEMDR community on Zenodo provides a coherent, searchable collection. The landing repository (*semdr-project/semdr-open-science*) aggregates all outputs under a single URL.
Accessible	Retrievable by identifier using standardised protocols; clear access conditions	Open access via GitHub HTTPS and Zenodo HTTPS download. Licensing (MIT for code, CC BY 4.0 for data) clearly stated in each repository. No authentication or registration required for access.
Interoperable	Use formal, accessible, shared languages and vocabularies	Open formats throughout: CSV for time-series and tabular data, JSON for structured data, IDF for EnergyPlus models (industry-standard open format). ISO 8601 timestamps. SI units for all measurements. Dublin Core metadata schema.
Reusable	Rich description, clear licence, provenance information	Each repository includes a comprehensive README with data dictionaries, collection methodology, quality notes, and known limitations. Clear provenance documented from source to published artefact. Permissive licences (MIT and CC BY 4.0) enable reuse with attribution.

Table 3 - FAIR Compliance Summary

5.2 Metadata Standards

All published outputs are accompanied by comprehensive metadata records following the Dublin Core Metadata Standard, supplemented with domain-specific metadata for energy and environmental data. Each record includes title, creator(s), description, subject keywords, creation date, temporal coverage, format, rights and licence information, related project identifier (COM-CONCEPT-ENERGY/0624/0160), and links to the SEMDR project community on Zenodo.

For energy time-series data specifically, additional metadata documents sensor and meter specifications and model numbers, measurement units (kWh for energy, kW for power, °C for temperature), sampling frequency and aggregation intervals, data collection period and any gaps, data processing and anonymisation procedures applied, and known quality issues or limitations. This comprehensive metadata supports effective data discovery and reuse by researchers and practitioners in the energy efficiency and demand response communities.

6 Open Science Infrastructure and Tools

This section describes the infrastructure and tools that support the SEMDR project's open science activities, including the GitHub organisation, the Zenodo repository, and the project website.

6.1 GitHub Repository

As described in Section 3.1, the SEMDR project maintains a dedicated GitHub organisation at https://github.com/semdr-project. The organisation serves as the authoritative home for all SEMDR open-source software and dataset outputs, providing version control, issue tracking, collaborative development, and integration with Zenodo for persistent citation. The organisation's landing repository offers a single entry point to the full set of project open science outputs.

6.2 Zenodo Repository

Zenodo, operated by CERN and funded through European Commission grants, serves as the primary long-term preservation platform for the SEMDR project's open science outputs. Zenodo provides automatic DOI assignment for each deposit via its GitHub integration, ensuring persistent and citable references, integration with the OpenAIRE infrastructure, which aggregates EU-funded research outputs and makes them discoverable across European research databases, support for versioned records, allowing deposits to be updated while maintaining links to previous versions, community features that enable the SEMDR project to organise all related outputs into a single discoverable collection, and long-term preservation commitment aligned with CERN's data archiving policies.

6.3 Project Website

The SEMDR project website (https://semdrcy.replit.app/) has been operational since M3 (June 2025) and serves as the primary public access point for project information, news, and deliverables. All public deliverables are available for download from the website. The website includes links to the SEMDR project GitHub organisation and to the corresponding Zenodo records, creating a cohesive open science platform that researchers and practitioners can navigate from any entry point to find any project output.

The project website will be maintained for a minimum of two years after project completion (until March 2028) to ensure continued accessibility to all published materials.

7 Open Science Outputs Inventory

Table 4 provides the consolidated inventory of all open science outputs produced or planned within the SEMDR project. This inventory encompasses software, datasets, publications, and infrastructure, providing a single reference point for the project's open science commitments and achievements.

ID	Type	Description	Licence	Status	Timeline
OS-1	Software	Cyprus electricity price forecaster (*semdr-project/cyprus-energy-market-prices-forecaster*). Day-ahead price forecasting tools. Zenodo DOI via GitHub integration.	MIT	**COMPLETED**	Published; live
OS-2	Software and dataset	Cyprus weather and electricity prices (*semdr-project/cyprus-weather-electricity-prices*). Combined dataset with acquisition scripts. Zenodo DOI via GitHub integration.	MIT / CC BY 4.0	**COMPLETED**	Published; live
OS-3	Dataset	Mediterranean hotel HVAC EnergyPlus baselines (*semdr-project/limassol-hotel-hvac-energyplus-dataset*). Five fixed-setpoint simulation baselines from the public DOE reference building. Zenodo DOI via GitHub integration.	CC BY 4.0	**COMPLETED**	Published; live
OS-4	Conference proceedings	CIGRE 2025 paper: "*Real-World Demonstration of Optimized Pool Pump Control for Demand Response in the Hospitality Sector.*" 32nd Scientific Conference of EU CIGRE, Athens, December 2025. Zenodo deposit with DOI planned.	-	PUBLISHED; Zenodo PLANNED	Zenodo within 6 months of proceedings availability
OS-5	Dataset	Anonymised pool pump production dataset (energy and water quality time-series with data dictionary and quality notes). Distinct from OS-3: real production data, not simulation.	CC BY 4.0	**COMMITTED**	By September 2026
OS-6	Journal paper	Methods and results paper (P-02) targeting Applied Energy or Energy and Buildings (Elsevier). Pre-print on Zenodo upon submission.	Open access	PLANNED	Q2 2026
OS-7	Dataset	Anonymised HVAC production dataset from the live deployment. Distinct from OS-3: real production data from the deployed algorithm.	CC BY 4.0	PLANNED	Post-project (following cooling season validation)
OS-8	Website	Project website: *https://semdrcy.replit.app*. Hosts non-confidential project information, news, and links to all open outputs.	Public	**OPERATIONAL**	Since M3
OS-9	Public deliverables	All deliverables classified as "Public" in the Description of Work, available through the project website.	Public	**COMPLETED**	-

Table 4 - Open Science Outputs Inventory

Of the nine identified outputs, six are already completed or operational at the time of writing: the three published open-source repositories (OS-1, OS-2, OS-3), the CIGRE conference paper (OS-4), the operational project website (OS-8), and the set of public deliverables available through the website (OS-9). OS-5 (the anonymised pool pump production dataset) carries a firm commitment for September 2026. The remaining outputs (OS-6 journal paper, OS-7 HVAC production dataset) are planned for the Q2 2026 and post-project periods respectively.

8 Key Performance Indicators and Timeline

This section presents the key performance indicators established for open science activities and assesses the project's progress against them.

8.1 Open Science KPIs

Table 5 summarises the key performance indicators for the SEMDR open science programme.

KPI	Target	Status	Timeline
Open-source repositories released	≥1 repository	**Achieved**	All live; Zenodo DOIs minted
Conference publications	≥1	**Achieved**	Published Dec 2025
CIGRE proceedings Zenodo deposit	1 deposit with DOI	Planned	Within 6 months of proceedings availability
Journal publications	≥1	In progress	Target: Q2 2026
Open access deliverables	All public deliverables	**Achieved**	Website since M3
Datasets deposited (Zenodo)	≥3 datasets	**Achieved**	OS-1, OS-2, OS-3 live; OS-5 by Sep 2026; OS-7 post-project
FAIR-compliant data plan	Full compliance	**Achieved**	Implemented for all live outputs
Project website active	Operational	**Achieved**	Active since M3 (June 2025)

Table 5 - Open Science Key Performance Indicators

8.2 Assessment

The SEMDR project has achieved six of its eight quantitative open science KPIs at the time of writing, with the remaining two (CIGRE Zenodo deposit and journal publication) on track for completion within their designated timelines. The open-source repository target of one has been exceeded with three repositories published. The dataset deposition target of three has been met, with a further two datasets committed or planned for post-project release.

The consortium's approach has been to prioritise the delivery of genuine, high-quality open science outputs over meeting abstract quantity targets. The Cyprus weather and electricity prices repository (OS-2) and the EnergyPlus baselines dataset (OS-3) were identified during the project as valuable open outputs that go beyond the minimum commitments in the Description of Work.

9 Conclusions and Future Activity

This section summarises the project's open science achievements to date and sets out the timeline for post-project activities.

9.1 Summary

The SEMDR project has developed and implemented a comprehensive open science strategy that aligns with the requirements of the Research and Innovation Foundation, the European Commission, and the Cyprus Open Science Practices framework. The strategy balances the goal of maximum openness with appropriate protection of commercial interests and operational privacy, following the principle of "as open as possible, as closed as necessary."

At the conclusion of the project period, the consortium has delivered the following open science outputs: three public open-source repositories hosted through a dedicated SEMDR project GitHub organisation, each with persistent Zenodo DOIs (OS-1 forecaster, OS-2 weather and electricity prices dataset, OS-3 EnergyPlus baselines); a conference paper at the 32nd EU CIGRE Scientific Conference (OS-4); an operational project website hosting all public deliverables (OS-8, OS-9); and a defined, committed pipeline for two further datasets and one journal paper.

The infrastructure necessary for post-project open science activities - including the GitHub organisation, Zenodo integration, data anonymisation procedures, and metadata templates - is established and operational. The project has therefore substantially fulfilled its open science commitments within the project period, with the remaining commitments carrying firm timelines for completion.

9.2 Post-Project Open Science Timeline

The following activities are scheduled for the post-project period:

- April-August 2026: Final data anonymisation and quality assurance for the pool pump production dataset (OS-5).
- By September 2026: Anonymised pool pump production dataset (OS-5) deposited on Zenodo with DOI.
- Q2 2026: Journal article (P-02 / OS-6) submitted to peer-reviewed journal, with pre-print deposited on Zenodo.
- Post-project (following cooling season validation): Anonymised HVAC production dataset (OS-7) deposited on Zenodo.
- Until March 2028: Continued maintenance of the project website, the SEMDR project GitHub organisation, and the Zenodo community.